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[> # Атака с помощью выбранного шифротекста-3
[> with(numtheory) :
[> with(RandomTools) :
[> p := nextprime(Generate(integer(range=2..10^10)));
                                     p := 581869333 (1)
[> q := nextprime(Generate(integer(range=2..10^10)));
                                     q := 4161255407 (2)
[> e := nextprime(Generate(integer(range=2..10^10)));
                                     e := 5244301331 (3)
[> n := p·q;
                                     n := 2421306908113733531 (4)
[> phi_n := (p - 1)·(q - 1);
                                     phi_n := 2421306903370608792 (5)
[> igcd(e, phi_n);
                                     1 (6)
[> d := 'd':
s := msolve(d·e = 1, phi_n) :
assign(s) :
d := d;
                                     d := 864117347993826155 (7)
[> # Ева
m1 := 645334539857 mod n;
m2 := 8967456326345 mod n;
m3 := m1·m2 mod n;
                                     m1 := 645334539857
                                     m2 := 8967456326345
                                     m3 := 1045915987081869080 (8)
[> # Алиса
M1 := m1&^d mod n;
M2 := m2&^d mod n;
M3 := M1·M2 mod n;
                                     M1 := 237458763302282677
                                     M2 := 2380569966618541017
                                     M3 := 206577519608622816 (9)
[> # Проверка
m := m3&^d mod n;
m - M3;
                                     m := 206577519608622816
                                     0 (10)
[>

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